



VP AIR TEST NEET (2025-26)

PART TEST - 04

DROPPER

DURATION: 180 Minutes

Date: 12-10-2025

M. MARKS: 720

ANSWER KEY

1. (3)	46. (4)	91. (4)	136. (1)
2. (2)	47. (1)	92. (2)	137. (3)
3. (4)	48. (1)	93. (3)	138. (1)
4. (3)	49. (2)	94. (2)	139. (3)
5. (4)	50. (2)	95. (2)	140. (4)
6. (2)	51. (3)	96. (2)	141. (4)
7. (2)	52. (3)	97. (4)	142. (3)
8. (2)	53. (1)	98. (2)	143. (2)
9. (2)	54. (3)	99. (2)	144. (4)
10. (4)	55. (2)	100. (1)	145. (2)
11. (2)	56. (3)	101. (4)	146. (1)
12. (2)	57. (3)	102. (2)	147. (3)
13. (1)	58. (2)	103. (3)	148. (4)
14. (4)	59. (1)	104. (4)	149. (3)
15. (1)	60. (3)	105. (1)	150. (3)
16. (3)	61. (4)	106. (4)	151. (3)
17. (2)	62. (1)	107. (1)	152. (2)
18. (2)	63. (2)	108. (4)	153. (3)
19. (4)	64. (4)	109. (3)	154. (2)
20. (1)	65. (4)	110. (4)	155. (2)
21. (4)	66. (4)	111. (3)	156. (3)
22. (2)	67. (1)	112. (4)	157. (2)
23. (2)	68. (2)	113. (4)	158. (3)
24. (1)	69. (4)	114. (2)	159. (2)
25. (1)	70. (2)	115. (3)	160. (3)
26. (2)	71. (3)	116. (3)	161. (3)
27. (4)	72. (1)	117. (1)	162. (3)
28. (3)	73. (3)	118. (2)	163. (2)
29. (2)	74. (2)	119. (3)	164. (3)
30. (4)	75. (2)	120. (2)	165. (1)
31. (2)	76. (2)	121. (3)	166. (1)
32. (3)	77. (2)	122. (3)	167. (1)
33. (4)	78. (1)	123. (2)	168. (1)
34. (4)	79. (3)	124. (3)	169. (1)
35. (3)	80. (3)	125. (3)	170. (3)
36. (1)	81. (1)	126. (2)	171. (2)
37. (1)	82. (3)	127. (3)	172. (2)
38. (2)	83. (3)	128. (4)	173. (1)
39. (4)	84. (4)	129. (3)	174. (1)
40. (3)	85. (4)	130. (2)	175. (1)
41. (4)	86. (1)	131. (4)	176. (2)
42. (4)	87. (1)	132. (1)	177. (3)
43. (3)	88. (2)	133. (1)	178. (1)
44. (1)	89. (1)	134. (2)	179. (1)
45. (1)	90. (2)	135. (3)	180. (3)

HINTS & SOLUTIONS

1. (3)

$$\therefore v = \omega r$$

since ω is same for both so, $v_1 = \omega r$

$$v_2 = \omega(2r)$$

$\frac{v_1}{v_2} = \frac{\omega r}{\omega(2r)} = \frac{1}{2}$; x and y are displacement of particle A and B is same time.

$$\Rightarrow \frac{x}{y} = \frac{1}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 104)

2. (2)

$$\text{As, } U = \frac{1}{2} FNL = \frac{F^2 L}{2AY} = \frac{F^2 L}{2\pi r^2 Y}$$

$$\Rightarrow U \propto \frac{L}{r^2} \quad (F \text{ and } Y \text{ are constant})$$

$$r_A = 2r_B$$

$$L_A = 3L_B \Rightarrow \frac{U_A}{U_B} = \left[\frac{L_A}{L_B} \right] \times \left[\frac{r_B}{r_A} \right]^2$$

$$= (3) \times \left(\frac{1}{2} \right)^2 = \frac{3}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 174)

3. (4)

Gravity at depth $d \left(= \frac{R}{2} \right)$

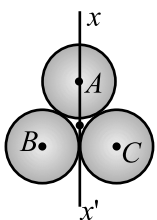
$$g' = g \left(1 - \frac{R/2}{R} \right) = \frac{g}{2} = 5 \text{ m/s}^2$$

$$\text{Weight at depth } \frac{R}{2} = \frac{200}{2} \times 5 = 100 \text{ N}$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

4. (3)

$$I = I_A + I_B + I_C$$



$$I_A = \frac{2}{5} MR^2$$

$$I_B = I_C = \frac{2}{5} MR^2 + MR^2 = \frac{7}{5} MR^2$$

$$\Rightarrow I = \frac{2}{5} MR^2 + \frac{7}{5} MR^2 + \frac{7}{5} MR^2;$$

$$I = \frac{16}{5} MR^2$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

5. (4)

Range of projectile $R = \frac{u^2 \sin 2\theta}{g}$ if u and θ are

constant then $R \propto \frac{1}{g}$

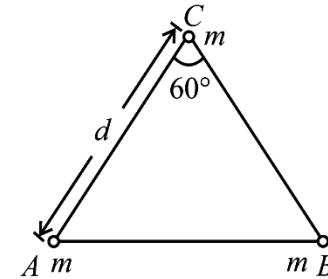
$$\frac{R_m}{R_e} = \frac{g_e}{g_m} \Rightarrow \frac{R_m}{R_e} = \frac{1}{0.2} \Rightarrow R_m = \frac{R_e}{0.2}$$

$$\Rightarrow R_m = 5R_e$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

6. (2)

$$U_a = U_{AB} + U_{BC} + U_{AC}$$



$$A = 3 \frac{Gmm}{d} = -3 \frac{Gm^2}{d}$$

$$\text{When } d \rightarrow 2d \Rightarrow U_b = -\frac{3Gm^2}{2d}$$

$$\text{Work done} = U_b - U_a = \frac{3Gm^2}{2d}$$

(NCERT Ed. 2025-26, 11th, Page No. 135)

7. (2)

$$V_0 = \sqrt{\frac{GM}{r}}$$

Where r is distance from the center of Earth.

$$\text{K.E.} \propto \frac{1}{r}$$

$$\frac{K_1}{K_2} = \frac{r_2}{r_1} = \frac{4R}{2R} = \frac{2}{1}$$

(NCERT Ed. 2025-26, 11th, Page No. 138)

8. (2)

Torque about O

$$\tau = F \times 40 + F \times 80 - (F \times 20 + F \times 60)$$

In clockwise direction

$$\tau = F \times 40$$

(NCERT Ed. 2025-26, 11th, Page No. 110)

9. (2)

So force on each columns will be

$$= \frac{mg}{4} = 125 \times 10^3 \text{ N.}$$

Thus Strain

$$\epsilon = \frac{F}{AY} = \frac{125 \times 10^3}{2.355 \times 2 \times 10^{11}} = 2.65 \times 10^{-7}$$

(NCERT Ed. 2025-26, 11th, Page No. 170)

10. (4)

$$\alpha = \frac{\omega}{t} = \frac{60}{5} = 12 \text{ rad/s}^2$$

$$\text{Angular disp. } \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$= \frac{1}{2} \times 12 \times 5^2 = 150 \text{ rad}$$

(NCERT Ed. 2025-26, 11th, Page No. 117)

11. (2)

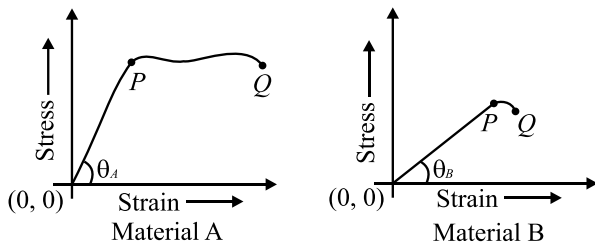
$$\text{Torque, } \tau = FR = I\alpha \Rightarrow 40 \times 0.1 = 0.4\alpha$$

$$\alpha = 10 \text{ rad/s}^2$$

$$\omega_f = 10 \times 10 = 100 \text{ rad/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

12. (2)



From the graph $\tan \theta_A > \tan \theta_B$

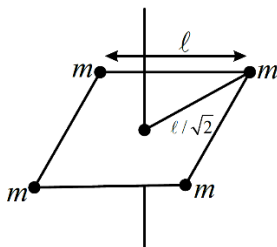
So material A has greater Young's modulus.

P and Q distance in material A is greater than P to Q distance in material B, which implies more plastic deformation is possible in A as compared to B.

(NCERT Ed. 2025-26, 11th, Page No. 170)

13. (1)

$$I = m \times \left(\frac{\ell}{\sqrt{2}} \right)^2 \times 4$$



$$\text{also, } I = MK^2$$

$$\text{where } M = 4m$$

$$\text{so, } m \times \frac{\ell^2}{2} \times 4 = 4m \times K^2$$

$$\therefore K = \frac{\ell}{\sqrt{2}}$$

(NCERT Ed. 2025-26, 11th, Page No. 115)

14. (4)

Gravitational field of ring

$$= \frac{Gmx}{(R^2 + x^2)^{\frac{3}{2}}}$$

Force between point mass and ring

$$\frac{GmM(\sqrt{8}R)}{(R^2 + 8R^2)^{\frac{3}{2}}} = \frac{GmM}{R^2} \times \frac{\sqrt{8}}{27}$$

(NCERT Ed. 2025-26, 11th, Page No. 130)

15. (1)

Angular momentum = Mass $\times$ Orbital Velocity $\times$ Radius

$$= m \times \left(\sqrt{\frac{GM}{R_0}} \right) \times R_0 = m\sqrt{GMR_0}$$

(NCERT Ed. 2025-26, 11th, Page No. 137)

16. (3)

Initial total energy = Initial kinetic energy + initial potential energy

$$= \frac{1}{2} m(0)^2 + \left(-\frac{GMm}{R_0} \right) = -\frac{GMm}{R_0}$$

Total energy, when it reaches the surface of earth

$$= \frac{1}{2} mv^2 + \left(-\frac{GMm}{R} \right)$$

Applying energy conservation,

$$\frac{1}{2} mv^2 - \frac{GMm}{R} = -\frac{GMm}{R_0}$$

$$v = \sqrt{2GM \left\{ \frac{1}{R} - \frac{1}{R_0} \right\}}$$

(NCERT Ed. 2025-26, 11th, Page No. 136)

17. (2)

$$I = 2[5(0.2)^2 + 2(0.4)^2]$$

$$= 1.04 \text{ kg m}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 114)

18. (2)

$$\int_0^v dV = -\int_\infty^x Edx$$

$$V = -\int_\infty^x \frac{2A}{x^3} dx$$

$$V = \frac{A}{x^2}$$

(NCERT Ed. 2025-26, 11th, Page No. 134)

19. (4)

Let, m be mass of satellite & M mass of earth

$$\text{Total energy} = \frac{-GMm}{2(2R)}$$

According to question $\frac{10^4 R}{6}$ J of energy is supplied to the satellite

$$\Rightarrow \frac{-GMm}{4R} + \frac{10^4 R}{6} = \frac{-GMm}{2r}$$

where r is new radius of revolving and

$$g = \frac{GM}{R^2}$$

$$-\frac{mgR}{4} + \frac{10^4 R}{6} = -\frac{mgR^2}{2r}$$

$$(m = 10^3 \text{ kg})$$

$$-\frac{10^3 \times 10 \times R}{4} + \frac{10^4 R}{6}$$

$$= -\frac{10^3 \times 10 \times R^2}{2r}$$

$$= -\frac{1}{4} + \frac{1}{6} = -\frac{R}{2r}$$

$$\therefore r = 6R$$

(NCERT Ed. 2025-26, 11th, Page No. 138)

20. (1)

Using conservation of angular momentum, we can write

$$v_1 = \frac{v_2 r_2}{r_1}$$

$$= \frac{8 \times 10^{10}}{1.6 \times 10^{12}} \times 6 \times 10^4$$

$$= 3 \times 10^3 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 129)

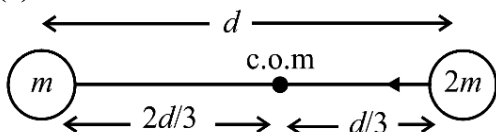
21. (4)

$$Y = \frac{F \times L}{A \times \Delta L} = \frac{500 \times 10 \times 0.7}{10 \times 10^{-6} \times 0.5 \times 10^{-2}}$$

$$= \frac{3500}{5 \times 10^{-8}} = 7 \times 10^{10} \frac{\text{N}}{\text{m}^2}$$

(NCERT Ed. 2025-26, 11th, Page No. 170)

22. (2)



$$F = \frac{G(2m)m}{d^2} = (2m)\omega^2 \left(\frac{d}{3} \right)$$

$$\frac{Gm}{d^2} = \omega^2 \frac{d}{3}$$

$$\Rightarrow \omega^2 = \frac{3Gm}{d^3} \Rightarrow \omega = \sqrt{\frac{3Gm}{d^3}}$$

$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{d^3}{3Gm}}$$

(NCERT Ed. 2025-26, 11th, Page No. 130)

23. (2)

Tension in wire-1 is double the tension in wire-2.

(NCERT Ed. 2025-26, 11th, Page No. 168)

24. (1)

Moment of inertia of a circular disc about centre of

mass and perpendicular to its plane $I_{cm} = \frac{mR^2}{2}$

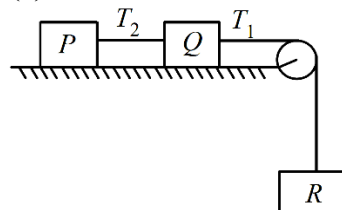
Using parallel axis theorem

$$I_{AB} = \frac{mR^2}{2} + m \left(\frac{2R}{3} \right)^2 = \frac{17}{18} mR^2$$

$$\therefore \frac{I_{AB}}{I_{cm}} = \frac{17}{9} \Rightarrow x = 17$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

25. (1)



Neglecting friction, the net force acting on each block is its mass multiplied by the common acceleration a .

The equations of motion for the system can be written as follows:

For block P:

$$T_2 = (3\text{kg}) \times a$$

For block Q:

$$T_1 - T_2 = (3\text{kg}) \times a$$

For block R:

$$(3\text{kg}) \times g - T_1 = (3\text{kg}) \times a$$

After solving above equation we get,

$$a = \frac{10}{3} \frac{\text{m}}{\text{s}^2}$$

$$T_1 = 20\text{N}$$

Now,

The longitudinal strain in wire B can be calculated using Hooke's Law:

$$\text{Strain} = \frac{\text{Longitudinal stress}}{\text{Young's modulus}}$$

$$\text{strain} = \frac{\text{stress}}{Y} = 2 \times 10^{-4}$$

(NCERT Ed. 2025-26, 11th, Page No. 170)

26. (2)

$$\text{Stress} = \frac{F}{A} = Y \frac{\Delta l}{l} \Rightarrow \Delta l = \frac{Fl}{AY}$$

$$\Delta l = \frac{200 \times 2}{2 \times 10^{-4} \times 10^{11}} = 2 \times 10^{-5} = 20 \mu\text{m}$$

(NCERT Ed. 2025-26, 11th, Page No. 170)

27. (4)

$$\frac{1}{2} I \omega^2 = 1000$$

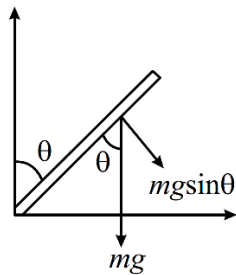
$$\omega = 10 \text{ rad/sec}$$

$$= \frac{10}{2\pi} \times 60 \text{ rpm}$$

$$= \frac{300}{\pi} \text{ rpm}$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

28. (3)



Taking torque about pivot $\tau = I\alpha$

$$mg \sin \theta \frac{l}{2} = \frac{ml^2}{3} \alpha$$

$$\alpha = \frac{3g}{2l} \sin \theta$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

29. (2)

Young's modulus is a property of the material, it does not depend on the dimensions.

(NCERT Ed. 2025-26, 11th, Page No. 170)

30. (4)

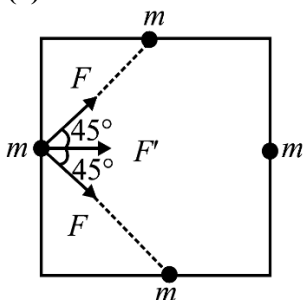
$$\vec{L} = \vec{r} \times m\vec{v}$$

$$L = (a)(mv_0)$$

Using right hand rule, the direction is along $(-\hat{k})$

(NCERT Ed. 2025-26, 11th, Page No. 105)

31. (2)



$$F_{net} = F' + 2F \cos 45^\circ$$

$$= \frac{Gm^2}{l^2} + \frac{2\sqrt{2}Gm^2}{l^2}$$

$$F_{net} = \frac{Gm^2}{l^2} (1 + 2\sqrt{2})$$

(NCERT Ed. 2025-26, 11th, Page No. 130)

32. (3)

$$\text{Elongation due to self weight} = \frac{MgL}{2AY}$$

(NCERT Ed. 2025-26, 11th, Page No. 170)

33. (4)

About centre of mass net torque is zero.

$$\tau_A = \tau_B$$

$$(T_A) \left(\frac{L}{10} \right) = T_B \left(\frac{L}{2} \right) \therefore \frac{T_A}{T_B} = \frac{5}{1}$$

(NCERT Ed. 2025-26, 11th, Page No. 111)

34. (4)

$$|\vec{E}_{\text{surface}}| = \frac{GM}{R^2}$$

$$15 \times 10^{-12} = \frac{6.7 \times 10^{-11} M}{(5.176)^2}$$

$$15 \times 10^{-12} = \frac{6.7 \times 10^{-11} M}{26.8}$$

$$M = 6 \text{ kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

35. (3)

$$L_i = I\omega_i = \frac{MR^2}{2} \cdot \omega = 100 \text{ kgm}^2/\text{s}$$

$$E_i = \frac{1}{2} \cdot \frac{MR^2}{2} \cdot \omega^2 = 500 \text{ J}$$

$$\vec{L}_i = \vec{L}_f \Rightarrow 100 = 2I\omega_f$$

$$\omega_f = 5 \text{ rad/sec}$$

$$E_f = 2 \times \frac{1}{2} \cdot \frac{5(2)^2}{2} \cdot (5)^2 = 250 \text{ J}$$

$$\Delta E = 250 \text{ J}$$

(NCERT Ed. 2025-26, 11th, Page No. 122)

36. (1)

$$\vec{v} = \frac{d\vec{r}}{dt} = \hat{i} + 6t\hat{j}$$

$$\vec{L} = m(\vec{r} \times \vec{v})$$

$$= (t\hat{i} + 3t^2\hat{j}) \times (\hat{i} + 6t\hat{j})$$

$$= 6t^2\hat{k} - 3t^2\hat{k}$$

$$= 3t^2\hat{k}$$

$$\vec{L}_{(t=1s)} = (3\hat{k}) \text{ kg m}^2\text{s}^{-1}$$

(NCERT Ed. 2025-26, 11th, Page No. 105)

37. (1)

$$|\vec{E}_1| = |\vec{E}_2|$$

$$\frac{G_1 M_1 x_1}{(R_1^2 + x_1^2)^{\frac{3}{2}}} = \frac{G M_2 x_2}{(R_2^2 + x_2^2)^{\frac{3}{2}}}$$

$$\frac{(R_2^2 + x_2^2)^{\frac{3}{2}}}{(R_1^2 + x_1^2)^{\frac{3}{2}}} = \frac{M_2 x_2}{M_1 x_1}$$

$$\frac{(5R^2)^{\frac{3}{2}}}{(5R^2)^{\frac{3}{2}}} = \frac{M_2}{M_1} \times \frac{1}{2}$$

$$\frac{M_1}{M_2} = \frac{1}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 130)

38. (2)

$$MOI = \frac{Mx^2}{12} \quad \dots(1)$$

$$MOI = MK^2 \quad \dots(2)$$

On comparing,

$$K = \frac{x}{\sqrt{12}}$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

39. (4)

M.O.I. about C.O.M. is Minimum

$$I = I_{C.M.} + Mx_0^2$$

$$I = 2x^2 - 12x + 27$$

$$\therefore \frac{dI}{dx} = 4x - 12 = 0 \Rightarrow x = 3$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

40. (3)

For shells *C* and *D*, the particle is inside the shell so force due to these shells is zero.

For shells *A* and *B* the particle is outside so these shells will apply some force on the particle.

(NCERT Ed. 2025-26, 11th, Page No. 131)

41. (4)

$$v_0 = \sqrt{\frac{GM}{r}}$$

(NCERT Ed. 2025-26, 11th, Page No. 137)

42. (4)

$$g_{up} = \frac{g}{\left(1 + \frac{r}{R_E}\right)^2}$$

$$g_{down} = g \left(1 - \frac{r}{R_E}\right)$$

$$\frac{g_{down}}{g_{up}} = \left(1 - \frac{r}{R_E}\right) \left(1 + \frac{r}{R_E}\right)^2$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

43. (3)

$$V = \pi r^2 L$$

$$\frac{\Delta V}{V} = \frac{dL}{L} - 2 \frac{dr}{r}$$

$$= \frac{dL}{L} - 2\sigma \frac{dL}{L}$$

$$\Rightarrow \frac{dV}{V} = \frac{F}{AY} (1 - 2\sigma)$$

$$\therefore \sigma = \frac{\frac{dr}{r}}{\frac{dL}{L}}$$

(NCERT Ed. 2025-26, 11th, Page No. 174)

44. (1)

$$\tau = I\alpha;$$

$$I' = (2m) \left(\frac{r}{2}\right)^2 = \frac{mr^2}{2} = \frac{I}{2}$$

$$\tau' = I'\alpha.$$

$$\frac{\tau'}{\tau} = \frac{I'\alpha}{I\alpha} = \frac{I'}{I} = \frac{1}{2}; \quad \tau' = \frac{\tau}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

45. (1)

Unit of angular acceleration is rad/s², Unit of torque is Newton-meter, Unit of angular momentum is kg m²/sec, unit of moment of inertia is kg m².

(NCERT Ed. 2025-26, 11th, Page No. 124)

46. (4)

- For every chemical reaction at equilibrium, ΔG of reaction is zero.
- The endothermic reactions are favoured at higher temperature and the exothermic reactions are favoured at lower temperature.

(NCERT Ed. 2025-26 11th Part-I Page No. 184, 187)

47. (1)

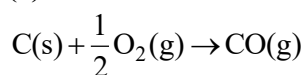
$$n = \frac{\Delta U}{C_V \Delta T} = \frac{4000}{12 \times 200} = 1.67$$

(NCERT Ed. 2025-26 11th Part-I Page No. 144)

48. (1)

At equilibrium, $r_f = r_b$ (NCERT Ed. 2025-26 11th Part-I Page No. 203)

49. (2)

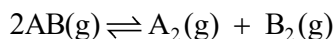


$$\Delta H - \Delta U = \Delta n_g RT = \frac{1}{2} \times 8.31 \times 298$$

$$= 1238.19 \text{ J} = 1.24 \text{ kJ}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 143)

50. (2)



- (i) 1 mol - -
 (f) 0.5 mol 0.25 mol 0.25 mol

$$K_c = \frac{(0.25)(0.25)}{(0.5)^2} = 0.25$$

(NCERT Ed. 2025-26 11th Part-I Page No. 206)

51. (3)

- $\left(\frac{dH}{dT}\right)_P = \frac{C_P dT}{dT} = C_P$ (for 1 mole)
- $\left(\frac{dU}{dT}\right)_V = \frac{C_V dT}{dT} = C_V$ (for 1 mole)
- $\left(\frac{dG}{dT}\right)_P = \left(\frac{VdP - SdT}{dT}\right)_P = -S$
- $\left(\frac{dG}{dP}\right)_T = \left(\frac{VdP - SdT}{dP}\right)_T = V$

(NCERT Ed. 2025-26 11th Part-I Page No. 145)

52. (3)

$$K_c = \frac{[CO_2]^3}{[CO]^3}$$

Addition or removal of $Fe_2O_3(s)$ and $Fe(l)$ will not affect K_c and hence equilibrium.

(NCERT Ed. 2025-26 11th Part-I Page No. 194)

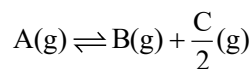
53. (1)

Isothermal process: $\Delta T = 0 \Rightarrow \Delta U = 0$ In vacuum, $P_{\text{ext}} = 0 \Rightarrow W = 0$ 1st law of thermodynamics,

$$\Delta U = q + w \quad \therefore q = 0$$

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54. (3)



$$(i) \quad 1 \quad - \quad -$$

$$(f) \quad 1 - \alpha \quad \alpha \quad \frac{\alpha}{2}$$

$$K_p = \frac{\left(\frac{P\alpha}{1 + \frac{\alpha}{2}}\right) \left(\frac{P\frac{\alpha}{2}}{1 + \frac{\alpha}{2}}\right)^{1/2}}{\left(\frac{P(1-\alpha)}{1 + \frac{\alpha}{2}}\right)}$$

$$K_p = \frac{\alpha}{1 - \alpha} \sqrt{\frac{P\alpha}{2\left(1 + \frac{\alpha}{2}\right)}}$$

But, $\alpha \ll 1$

$$\therefore 1 - \alpha \approx 1 \text{ and } 1 + \frac{\alpha}{2} \approx 1$$

$$\therefore K_p = \frac{\alpha}{1} \sqrt{\frac{P\alpha}{2}} = \frac{P^{1/2} \alpha^{3/2}}{2^{1/2}}$$

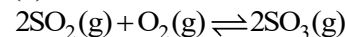
(NCERT Ed. 2025-26 11th Part-I Page No. 180)

55. (2)

Molarity, temperature, refractive index and specific heat capacity, molarity of a system do not depend on size/mass of the system hence these are intensive properties.

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56. (3)



$$K = \frac{[SO_3]^2}{[SO_2]^2 [O_2]} = \frac{1}{\left(\frac{[SO_2][O_2]^{1/2}}{[SO_3]}\right)^2}$$

$$K = \frac{1}{K_c^2} = \frac{1}{(6.4 \times 10^{-2})^2}$$

$$K = \frac{10^4}{6.4 \times 6.4} = 244.14$$

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57. (3)

A system at Higher temperature has greater randomness in it than one at lower temperature. Thus, temperature is the measure of average chaotic motion of particles in the system.

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75. (2)

$$T = \frac{\Delta H}{\Delta S} = \frac{40 \times 1000}{60} = 666.67 \text{ K}$$

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76. (2)

$$\text{XY}_2(\text{s}) \rightleftharpoons \underset{\text{p}}{\text{X}}(\text{g}) + \underset{2\text{p}}{2\text{Y}}(\text{g})$$

$$p_{\text{total}} = p + 2p = 3p = 12$$

$$\therefore p = 4 \text{ atm}$$

$$K_p = (p)^1 (2p)^2 = 4p^3 = 4(4)^3 = 256 \text{ atm}^3$$

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77. (2)

For a diatomic gas: $\frac{C_p}{C_v} = 1.4$

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78. (1)

$$\text{Unit of } K_c = \left(\frac{\text{mol}}{\text{L}} \right)^{\Delta n_g} = \left(\frac{\text{mol}}{\text{L}} \right)^{(2+3) - \left(2 + \frac{5}{2} \right)}$$

$$= \left(\frac{\text{mol}}{\text{L}} \right)^{1/2} = \text{mol}^{1/2} \text{ L}^{-1/2}$$

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79. (3)

On applying $2 \times \text{eq}(1) + \text{eq}(2) - \text{eq}(3)$

$$\text{B} + \text{D} \rightarrow \text{E} + 2\text{C},$$

$$\therefore \Delta H = 2(100) + (-120) - (300) = -220 \text{ kJ mol}^{-1}$$

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80. (3)

$$\Delta n_g = (1) - \left(1 + \frac{1}{2} \right) = -\frac{1}{2}$$

$$K_p = K_c (RT)^{-\frac{1}{2}} \therefore \frac{K_c}{K_p} = \sqrt{RT}$$

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81. (1)

$$\Delta_r H = \Sigma(\text{BE})_R - \Sigma(\text{BE})_P$$

$$= (4\text{BE}_{\text{C-H}} + \text{BE}_{\text{C=C}} + \text{BE}_{\text{H-H}}) - (6\text{BE}_{\text{C-H}} + \text{BE}_{\text{C-C}})$$

$$= (4z + y + w) - (6z + x) = y + w - 2z - x \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 154)

82. (3)

When an inert gas is added to the equilibrium, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ at constant pressure, the equilibrium will not disturb because in the reaction, $\Delta n_g = 0$.

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83. (3)

$$\text{H}_2\text{O}(\ell) \rightarrow \text{H}_2\text{O}(\text{g})$$

$$\Delta U = \Delta H - \Delta n_g RT$$

$$= 41 - 1 \times 8.3 \times 10^{-3} \times 373 = 37.9 \text{ kJ mol}^{-1}$$

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84. (4)

- At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is not increased at melting point.
- A catalyst can not alter the equilibrium constant (K_c) of the reaction at constant temperature.

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85. (4)

$$W = -2.303 nRT \log \left(\frac{V_2}{V_1} \right)$$

$$W = -2.303 \times 2 \times 8.314 \times 300 \times \log(2)$$

$$= -3457.97 \text{ J} = -3.46 \text{ kJ}$$

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86. (1)

$$\Delta G^\circ = -2.303 RT \log K$$

$$= -2.303 \times R \times 298 \times \log 10^{-14}$$

$$= 2.303 \times 14 \times R \times 298 \text{ J mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 184)

87. (1)

$$\Delta U = q + w = -1 + 5 = 4 \text{ kJ}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 140)

88. (2)

- Endothermic process on increasing temperature moves in forward direction.
- On increasing pressure, solubility of gas in water increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

89. (1)

Third law of thermodynamics:
The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero.

(NCERT Ed. 2025-26 11th Part-I Page No. 161)

90. (2)

$$K_p = (p_{\text{O}_2})^{1/2}$$

$$9 = (p_{\text{O}_2})^{1/2}$$

$$p_{\text{O}_2} = 81 \text{ atm}$$

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91. (4)

The discontinuously synthesised fragments (during replication) are later joined by the enzyme DNA ligase. Faulty or non-functional DNA ligase will not be able to do it, leading to accumulation of DNA fragments. Transcription is performed by RNA polymerase, it will not be impacted.

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92. (2)

Non-living objects also grow if we take increase in body mass as a criterion for growth. Mountains, boulders and sand mounds do grow. However, this kind of growth exhibited by non-living objects is by accumulation of material on the surface. Metabolism is a defining feature of all living organisms without exception. Consciousness is the defining property of living organisms. Perhaps, the most obvious and technically complicated feature of all living organisms is this ability to sense their surroundings or environment and respond to these environmental stimuli which could be physical, chemical or biological. Non-living objects do not show this property.

(NCERT Ed. 2025-26, 11th, Page No. 4)

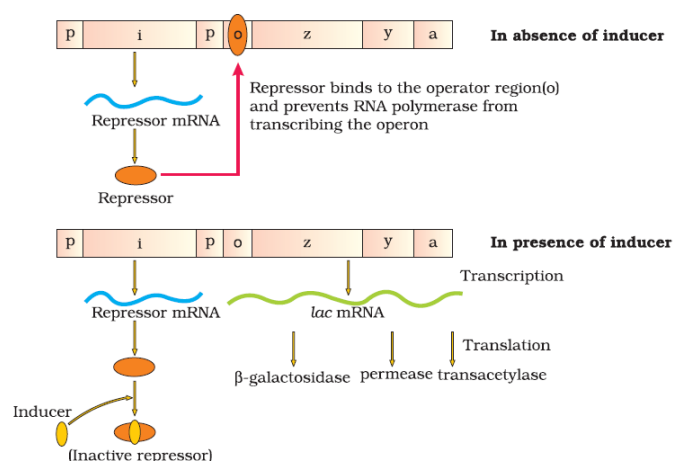
93. (3)

The backbone of a polynucleotide chain is formed due to sugar and phosphates. The nitrogenous bases linked to sugar moiety project inside from the backbone. The phosphate groups are bound to the pentose sugars. The bond between the pentose sugar and the nitrogenous base is called a N-glycosidic linkage. The phosphoester linkage is between the phosphate group and the sugar.

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94. (2)

In *lac* operon (here *lac* refers to lactose), a polycistronic structural gene is regulated by a common promoter and regulatory genes. The *lac* operon consists of one regulatory gene (the *i* gene – here the term *i* does not refer to inducer, rather it is derived from the word inhibitor) and three structural genes (*z*, *y*, and *a*). The *i* gene codes for the repressor of the *lac* operon. The *z* gene codes for beta-galactosidase (β -gal), which is primarily responsible for the hydrolysis of the disaccharide, lactose into its monomeric units, galactose and glucose. Regulation of *lac* operon by repressor is referred to as negative regulation.

The *lac* Operon

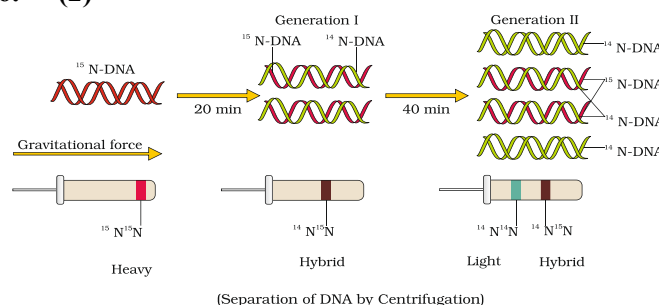
(NCERT Ed. 2025-26, 12th, Page No. 94)

95. (2)

- Triticum* - Genus
- Anacardiaceae - Family
- Poales - Order
- Angiospermae - Division

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96. (2)



(Separation of DNA by Centrifugation)

Meselson and Stahl's Experiment

In Meselson and Stahl's experiment, the DNA that was extracted from the culture one generation after the transfer from ^{15}N to ^{14}N medium [that is after 20 minutes; *E. coli* divides in 20 minutes] had a hybrid or intermediate density. ^{15}N is not a radioactive isotope, and it can be separated from ^{14}N only based on densities. The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase. They grew some viruses on a medium that contained radioactive phosphorus and some others on medium that contained radioactive sulfur.

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97. (4)

Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin.

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98. (2)
The fact that the code is a triplet explains how 4 bases can specify all amino acids ($4^3 = 64$ possible combinations). It describes the structure of the code. The nearly universal nature of code implies that a particular codon codes for same amino acid in almost all living organisms.

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99. (2)
An isolated metabolic reaction(s) outside the body of an organism, performed in a test tube is neither living nor non-living. Hence, while metabolism is a defining feature of all living organisms without exception, isolated metabolic reactions *in vitro* are not living things but surely living reactions.

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100. (1)
The steps of mRNA synthesis are:
1. Formation of heterogenous nuclear RNA (hnRNA) in the nucleus.
2. hnRNA undergoes additional processing called as capping and tailing. In capping an unusual nucleotide (methyl guanosine triphosphate) is added to the 5'-end of hnRNA.
3. In tailing, adenylate residues (200-300) are added at 3'-end
4. It is subjected to a process called splicing where the introns are removed and exons are joined.
5. It is the fully processed hnRNA, now called mRNA, that is transported out of the nucleus for translation.

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101. (4)
The steps of DNA fingerprinting included (i) isolation of DNA, (ii) digestion of DNA by restriction endonucleases, (iii) separation of DNA fragments by electrophoresis, (iv) transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon, (v) hybridisation using labelled VNTR probe, and (vi) detection of hybridised DNA fragments by autoradiography.

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102. (2)
Musca and *Homo* – Present in same kingdom i.e. Animalia
Mangifera and *Datura* – Members of family Anacardiaceae and Solanaceae, respectively
Solanum and *sapiens* – Genus and species, respectively

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103. (3)
The bacterial cell treats the viral genetic material as if it was its own and subsequently manufactures more virus particles.

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104. (4)
Species is the lowest category of the taxonomical hierarchy. The taxonomical hierarchy consists of 7 broad categories which are further divided into sub-categories. As we go higher from species to kingdom, the number of common characteristics goes on decreasing.

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105. (1)
Some of the salient observations drawn from human genome project are as follows: The human genome contains 3164.7 million bp. Less than 2 per cent of the genome codes for proteins. The average gene consists of 3000 bases. Chromosome 1 has most genes (2968), and the Y has the fewest (231). Repeated sequences make up very large portion of the human genome.

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106. (4)
All species in a genus belong to the same family, and all genera in a family belong to the same order. However, a single family can contain multiple, distinct genera. The basic requirement to place an organism in various categories is the knowledge of characters of an individual or group of organisms.

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107. (1)
In a transcription unit, the promoter is a DNA sequence that provides binding site for RNA polymerase. The strand that has the polarity $3' \rightarrow 5'$ acts as a template. The terminator is located towards 3'-end and it usually defines the end of the process of transcription. The other strand which has the polarity ($5' \rightarrow 3'$) and the sequence same as RNA (except thymine at the place of uracil), which does not code for anything, is referred to as coding strand.

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108. (4)
Caenorhabditis elegans (a free living non-pathogenic nematode) belongs to kingdom Animalia. *Escherichia coli* and *Streptococcus pneumoniae* are bacteria, which belong to the Kingdom Monera.

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109. (3)

James Dewey Watson was born in Chicago on 6 April 1928. In 1947, he received B.Sc. degree in Zoology. During these years his interest in bird-watching had matured into a serious desire to learn genetics. This became possible when he received a Fellowship for graduate study in Zoology at Indiana University, Bloomington, where he received his Ph.D. degree in 1950 on a study of the effect of hard X-rays on bacteriophage multiplication. He met Crick and discovered their common interest in solving the DNA structure. Their first serious effort, was unsatisfactory. Their second effort based upon more experimental evidence and better appreciation of the nucleic acid literature, resulted, early in March 1953, in the proposal of the complementary double-helical configuration.

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110. (4)

Genus (C) is a group of closely related species (D). Therefore, a genus can contain multiple species, but a species can only belong to one genus. The animals classified in phylum 'A' (Chordata) possess notochord and dorsal hollow neural system. Monkey and Gorilla fall in category 'B' (Order) named Primata.

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111. (3)

In eukaryotes, there is a set of positively charged, basic proteins called histones. A protein acquires charge depending upon the abundance of amino acids residues with charged side chains. Histones are rich in the basic amino acid residues lysine and arginine. Both the amino acid residues carry positive charges in their side chains. Histones are organised to form a unit of eight molecules called histone octamer. The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome. A typical nucleosome contains 200 bp of DNA helix. Nucleosomes constitute the repeating unit of a structure in nucleus called chromatin, thread-like stained (coloured) bodies seen in nucleus.

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112. (4)

Similarly, 'animals', 'mammals', 'dogs' are all taxa – but a dog is a mammal and mammals are animals. Therefore, 'animals', 'mammals' and 'dogs' represent taxa at different levels.

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113. (4)

The ribosome acts as a catalyst (23S rRNA in bacteria is the enzyme- ribozyme) for the formation of peptide bond. When the small subunit encounters an mRNA, the process of translation of the mRNA to protein begins. In the first phase itself amino acids are activated in the presence of ATP and linked to their cognate tRNA—a process commonly called as charging of tRNA or aminoacylation of tRNA to be more specific. An mRNA also has some additional sequences that are not translated and are referred as untranslated regions (UTR). The UTRs are present at both 5'-end (before start codon) and at 3'-end (after stop codon). They are required for efficient translation process. In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed. Consequently, the transcription and translation can be coupled in bacteria.

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114. (2)

Properties of tissues are not present in the constituent cells but arise as a result of interactions among the constituent cells. Similarly, properties of cellular organelles are not present in the molecular constituents of the organelle but arise as a result of interactions among the molecular components comprising the organelle.

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115. (3)

Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion. Such mutations are referred to as frame-shift insertion or deletion mutations. The total length of the gene is now shorter by one base, which will change the total number of codons. A frameshift mutation almost always leads to a drastically different protein.

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116. (3)

Ernst Mayr, the Harvard University evolutionary biologist has been called 'The Darwin of the 20th century'.

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117. (1)

The 2'-OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable. DNA chemically is less reactive and structurally more stable when compared to RNA. Therefore, among the two nucleic acids, the DNA is a better genetic material.

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118. (2)

In lower organisms like yeast and hydra, we observe budding. In *Planaria* (flat worms), we observe true regeneration, i.e., a fragmented organism regenerates the lost part of its body and becomes a new organism. The fungi, the filamentous algae, the protonema of mosses, all easily multiply by fragmentation. Further, there are many organisms which do not reproduce (mules, sterile worker bees, infertile human couples, etc).

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119. (3)

Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting. All the genes that are expressed as RNA are referred to as Expressed Sequence Tags (ESTs). The VNTR belongs to a class of satellite DNA referred to as mini-satellite. Restriction digestion refers to cleavage of DNA by restriction endonucleases at specific sites.

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120. (2)

S strain → Inject into mice → Mice die
R strain → Inject into mice → Mice live
S strain (heat-killed) → Inject into mice → Mice live
S strain (heat-killed) + R strain (live) → Inject into mice → Mice die

Injection of heat-killed R strain pneumococcus: While not a standard part of the experiment, since the live R strain is already harmless, the heat-killed version would also be harmless. Mice would live.

(NCERT Ed. 2025-26, 12th, Page No. 78)

121. (3)

All animals belonging to various phyla are assigned to the highest category called Kingdom Animalia in the classification system of animals.

(NCERT Ed. 2025-26, 11th, Page No. 8)

122. (3)

The bases in two DNA strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds.

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123. (2)

Members of the family Canidae, which includes dogs, belong to the class Mammalia. The class Insecta is for insects. The family Canidae is in the order Carnivora. The genus *Petunia* belongs to the family Solanaceae and the order Polymoniales. These are entirely different taxonomic orders. The genus *Felis* includes small cats and belongs to the family Felidae. While both the family Canidae and the family Felidae belong to the same order, Carnivora, they are not "exactly similar." Members of the family Canidae are animals and are classified in the kingdom Animalia. Wheat, being a plant, is classified in the kingdom Plantae.

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124. (3)

AUG has dual functions. It codes for Methionine (met), and it also act as initiator codon. UAA, UAG, UGA are stop terminator codons. GUG codes for Valine.

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125. (3)

DNA is a long polymer of deoxyribonucleotides. The length of DNA is usually defined as number of nucleotides (or a pair of nucleotide referred to as base pairs) present in it. This also is the characteristic of an organism. For example, Bacteriophage lambda has 48502 base pairs (bp), *Escherichia coli* has 4.6×10^6 bp, and haploid content of human DNA is 3.3×10^9 bp. Genetic material in QB bacteriophage is RNA.

(NCERT Ed. 2025-26, 12th, Page No. 77)

126. (2)

In the taxonomical hierarchy, the category Sapindales is placed between Dicotyledonae and Anacardiaceae.

(NCERT Ed. 2022-23, 11th, Page No. 11)

127. (3)

The ribosome consists of structural RNAs and about 80 different proteins. Both DNA and RNA are able to mutate. In RNA the uracil is found at the place of thymine (5-methyl uracil, another chemical name for thymine). DNA has evolved from RNA with chemical modifications that make it more stable.

(NCERT Ed. 2025-26, 12th, Page No. 86)

128. (4)

Early man deified some of the inanimate matter (wind, sea, fire etc.) and some among the animals and plants.

(NCERT Ed. 2022-23, 11th, Page No. 3)

129. (3)

RNA synthesis, or transcription, is a continuous process. Unlike DNA replication, there is no "lagging strand" in transcription that requires discontinuous synthesis.

(NCERT Ed. 2025-26, 12th, Page No. 82)

130. (2)

In order to facilitate the study, number of scientists have established procedures to assign a scientific name to each known organism. This is acceptable to biologists all over the world. For plants, scientific names are based on agreed principles and criteria, which are provided in International Code for Botanical Nomenclature (ICBN).

Animal taxonomists have evolved International Code of Zoological Nomenclature (ICZN). The scientific names ensure that each organism has only one name. Description of any organism should enable the people (in any part of the world) to arrive at the same name. They also ensure that such a name has not been used for any other known organism.

(NCERT Ed. 2025-26, 11th, Page No. 6)

131. (4)

Soluble RNA is another name for tRNA (transfer RNA). RNA that gets translated is mRNA (messenger RNA). mRNA carries the genetic code from the DNA in the nucleus to the ribosomes in the cytoplasm, where this code is translated into a protein sequence. The ribosome also acts as a catalyst (23S rRNA in bacteria is the enzyme-ribozyme) for the formation of peptide bond. RNA transcribed by RNA polymerase I in nucleus refers to the synthesis of most ribosomal RNAs in eukaryotes. RNA polymerase I is specifically responsible for transcribing the precursor rRNA, which is then processed into the 18S, 5.8S, and 28S rRNAs.

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132. (1)

Polymorphism (variation at genetic level) arises due to mutations. Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting. If an inheritable mutation is observed in a population at high frequency, it is referred to as DNA polymorphism. The probability of such variation to be observed in noncoding DNA sequence would be higher as mutations in these sequences may not have any immediate effect/impact in an individual's reproductive ability. As the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing, in case of disputes.

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133. (1)

The two RNA molecules if produced simultaneously would be complementary to each other, hence would form a double stranded RNA. This would prevent RNA from being translated into protein and the exercise of transcription would become a futile one.

(NCERT Ed. 2025-26, 12th, Page No. 87)

134. (2)

The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope (EM). A typical nucleosome contains 200 bp of DNA helix. Therefore, with 50 beads observed, the total length of DNA would be roughly $50 \text{ nucleosomes} \times 200 \text{ bp/nucleosome} = 10,000 \text{ bp}$. The DNA that forms chromatin and chromosomes in the nucleus is a double-stranded helix. This double helix is what wraps around the histone proteins to form the nucleosomes.

(NCERT Ed. 2025-26, 12th, Page No. 85)

135. (3)

The pitch of the DNA helix is 3.4 nm and there are roughly 10 bp in each turn. Consequently, the distance between a bp in a helix is approximately 0.34 nm. The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds. As a result, always a purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix.

(NCERT Ed. 2025-26, 12th, Page No. 80)

136. (1)

The ureters enter the bladder at an angle, so when the bladder wall contracts, it squeezes the openings of the ureters shut. This acts like a valve and stops urine from going back up to the kidneys.

Urination (micturition) requires the relaxation of the urethral sphincters to allow urine to pass out of the body. Contraction of these sphincters is what holds the urine in.

(NCERT Ed. 2025-26, 11th, Page No. 213)

137. (3)

All three major nitrogenous waste products—ammonia, urea, and uric acid—are ultimately derived from the catabolism of proteins and nucleic acids.

The water requirement for excretion varies significantly. Ammonia is the most toxic and requires a large amount of water for its excretion. Urea is less toxic and requires a moderate amount of water. Uric acid is the least toxic and can be excreted with minimal water loss.

The toxicity levels are not uniform. Ammonia is highly toxic, urea is moderately toxic, and uric acid is the least toxic. This difference in toxicity is a key factor in the evolutionary adaptation of different animals to different environments.

In humans, the synthesis of urea from ammonia occurs primarily in the liver via the urea cycle, not in the kidney nephrons. The kidneys are responsible for filtering the urea from the blood and excreting it in urine.

(NCERT Ed. 2025-26, 11th, Page No. 205)

138. (1)

Approximately 85% of nephrons in the human kidney are cortical nephrons. They have short loops of Henle that barely extend into the renal medulla. These nephrons are responsible for the primary and bulk reabsorption and secretion processes that lead to the formation of urine.

The high concentration of solutes in the medullary interstitium is maintained by the countercurrent mechanism, and urea recycling is a crucial part of this. However, urea is transported out of the collecting duct and re-enters the interstitium to maintain its high osmolarity. It does not re-enter the *vasa recta* (the blood vessels surrounding the loop of Henle) in this process; rather, some of it diffuses into the thin ascending limb of the loop of Henle and re-enters the tubule fluid, completing the recycling loop.

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139. (3)

Ascending limb of the Loop of Henle is known as the "diluting segment" because the filtrate's osmolarity decreases significantly as it passes through it (from hypertonic at the bottom of the loop to hypotonic as it enters the DCT). The key physiological property of the thick ascending limb is that it is impermeable to water but contains pumps that actively transport NaCl (and other electrolytes) out of the tubule.

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140. (4)

List-I	List-II
Mammals	Kidney
<i>Amphioxus</i>	Protonephridia
Malpighian tubules	Cockroaches
Antennal glands	Crustaceans

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141. (4)

- A - Liver: The process begins with Angiotensinogen, which is a precursor protein that is continuously produced and released into the bloodstream by the liver.
- B - Kidney: When blood pressure drops or fluid volume is low, specialized cells in the kidney detect this change and release an enzyme called Renin.
- Renin acts on Angiotensinogen, converting it into Angiotensin I. (Note: Angiotensin I is then converted to Angiotensin II by another enzyme, ACE, which is not shown in this simplified chart. Angiotensin II is the active form that causes the stimulation).
- Angiotensin (specifically Angiotensin II) stimulates the adrenal cortex, which is the outer part of the adrenal gland.
- C - Aldosterone: In response to this stimulation, the adrenal cortex secretes the hormone Aldosterone. Aldosterone then acts on the kidneys to increase the reabsorption of sodium and water, which helps to increase blood volume and, consequently, raise blood pressure back to normal.

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142. (3)

Glomerular filtration is relatively non-selective (filters water and small solutes but not large proteins or cells) and is driven by the glomerular capillary blood pressure (hydrostatic pressure).

The kidney has an inner concave surface and an outer convex surface.

The renal tubule of a nephron consists of the Bowman's capsule, PCT, Loop of Henle, and DCT. The collecting duct has a different embryological origin and is structurally separate, although it is functionally connected.

The initial filtrate has an osmolarity similar to blood plasma (~300 mOsm/L). The human kidney can produce urine with a maximum concentration of ~1200 mOsm/L, which is four times the concentration of the initial filtrate.

In most fishes (which are ammonotelic), the majority of ammonia is excreted directly into the water by diffusion across the gills, not through the kidneys. The kidneys are more involved in osmoregulation.

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143. (2)

- The Loop of Henle is called the counter-current multiplier because it uses energy (active transport of NaCl out of the ascending limb) to create and amplify (multiply) the osmotic gradient in the renal medulla, making it progressively saltier from the cortex to the deep medulla.
- The *vasa recta* is called the counter-current exchanger. Its primary role is to supply oxygen and nutrients to the medullary tissue without disrupting the high solute gradient created by the Loop of Henle. As blood flows down the descending limb of the *vasa recta*, it passively loses water and gains solutes. As it flows up the ascending limb, it gains water and loses solutes. This passive exchange ensures the gradient is maintained, not washed out.

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144. (4)

List-I	List-II
Sweat gland	Eliminates urea, salts, and water through skin surface
Excess ingestion	Leads to accumulation of ions like Na ⁺ , K ⁺ , Cl ⁻ , phosphate, sulphate, etc., in animals.
Ammonotelism	Process of excreting ammonia
Uric acid	Most water-efficient excretory waste

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145. (2)

The adult human kidney typically measures 10-12 cm in length and 5-7 cm in width, which is correct. However, it is located on the dorsal body wall, not the ventral. The kidneys are situated on either side of the vertebral column, well behind the peritoneal lining of the abdominal cavity.

Defecation is the process of eliminating solid waste from the body. This waste, known as feces, is primarily composed of undigested food, fiber, bacteria, and water that were never absorbed into the body's metabolic pathways. True physiological excretion involves the removal of metabolic waste products, such as urea, from the bloodstream, which is performed by organs like the kidneys and liver. Thus, defecation is more accurately described as elimination rather than excretion.

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146. (1)

The correct logical sequence for the steps of haemodialysis is (1) C → B → A.

- **C.** The first step is to draw blood from an artery in the patient. To prevent this blood from clotting within the artificial kidney (the dialyzer), an anticoagulant like heparin is added.
- **B.** Next, the heparinized blood is passed through the dialyzer. Inside, the blood flows alongside a specially formulated dialysis fluid. Due to a concentration gradient, waste products like urea and excess salts diffuse from the blood into the fluid, while essential substances are retained in the blood.
- **A.** Finally, after the blood has been dialyzed, it is no longer needed to be protected from clotting. Therefore, an anti-heparin is added to neutralize the heparin. The cleaned and restored blood is then infused back into the patient's body through a vein.

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147. (3)

The kidneys are responsible for filtering and excreting urea from the blood. In a patient with renal failure, the kidneys are already non-functional, which is why urea accumulates to high levels in the first place.

The liver is the sole site of the urea cycle, where ammonia is converted into urea.

When the patient, who already cannot excrete urea (due to renal failure), also suffers from acute liver failure, the production of new urea will cease.

Since no new urea is being produced (liver failure) and the existing urea cannot be removed (renal failure), the concentration of urea in the blood will stop rising and remain at the high level it was at when the liver failed.

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148. (4)

The human lungs excrete approximately 200 mL of CO₂ per minute at rest.

The normal Glomerular Filtration Rate (GFR) in a healthy adult is approximately 125 mL per minute (which equals 180 Liters per day).

A healthy adult on a normal diet excretes about 25-30 grams of urea per day.

Each human kidney contains nearly one million (1,000,000) nephrons, not one billion.

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149. (3)

The long loop of Henle and the parallel *vasa recta* are crucial for establishing and maintaining the osmotic gradient in the renal medulla. This gradient is essential for the kidney's ability to produce highly concentrated urine and conserve water.

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150. (3)

It is a fundamental clinical and physiological fact that the urine of a healthy person is free of glucose. The presence of glucose in urine (glycosuria) is a key indicator of a pathological condition, most commonly diabetes mellitus.

The process of urine formation begins with non-selective glomerular filtration, where small molecules like glucose, amino acids, and salts pass from the blood into the filtrate. The cells lining the proximal convoluted tubule (PCT) have specialized transporters that use energy (active transport) to move all of this filtered glucose out of the tubule and back into the bloodstream.

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151. (3)

- The entire micturition reflex is initiated by the stimulation of stretch receptors as the bladder fills with urine.
- These receptors send an afferent (sensory) signal to the central nervous system (spinal cord), which in turn creates the conscious sensation of needing to urinate and triggers the reflex motor response (contraction of the detrusor muscle).
- If these receptors are removed or non-functional, the CNS never receives the signal that the bladder is full.
- Therefore, the individual will not feel the urge to urinate, and the reflex arc that causes the bladder to contract will not be activated. This leads to the bladder overfilling, a condition known as urinary retention.

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152. (2)

When a person drinks a lot of water, their blood osmolarity decreases (it becomes less concentrated). Osmoreceptors in the hypothalamus detect this change and are less stimulated. This, in turn, reduces the release of Antidiuretic Hormone (ADH) from the posterior pituitary, leading to less water reabsorption and an increase in urine output.

Cold exposure leads to peripheral vasoconstriction to conserve heat. This shunts blood away from the skin and into the body's core, increasing central blood volume. This increase is detected by stretch receptors in the atria and great vessels, which signal for a decrease in ADH release. The reduction in ADH leads to increased urination, a phenomenon known as cold diuresis.

During summer, heavy water loss through sweating and evaporation causes the blood to become more concentrated (increased osmolarity). This change is detected by osmoreceptors, which stimulate the release of ADH, not suppress it. ADH then helps the body conserve water by promoting reabsorption in the kidneys.

The name "antidiuretic hormone" refers to its function in reducing urination. However, ADH is also known as vasopressin because it causes constriction of blood vessels (vasoconstriction), not dilation. This action helps to raise blood pressure.

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153. (3)

The descending limb of the loop of Henle is permeable to water but impermeable to electrolytes. As the filtrate moves down into the hypertonic medullary interstitium, water moves out via osmosis, making the filtrate progressively more concentrated (hypertonic).

The ascending limb is impermeable to water. It actively and passively transports electrolytes, primarily NaCl, from the filtrate into the medullary interstitium. This process is crucial for establishing and maintaining the high osmotic gradient in the medulla.

As the filtrate moves up the ascending limb, it becomes progressively more dilute, not more concentrated. This is because electrolytes are being removed from the filtrate and pumped into the medullary fluid, while water cannot follow.

Reabsorption is minimum in its ascending limb.

However, this region plays a significant role in the maintenance of high osmolarity of medullary interstitial fluid.

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154. (2)

The renal cortex is the outer part of the kidney, and the renal medulla is the inner part, which is made up of cone-shaped structures called renal pyramids. The Columns of Bertini are simply the parts of the cortex that extend down into the space between these pyramids.

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155. (2)

When blood pressure is high, the walls of the heart's atria are stretched. This stretch triggers the release of Atrial Natriuretic Factor (ANF). ANF acts as a powerful vasodilator, widening blood vessels to decrease peripheral resistance. It also inhibits the release of renin from the kidneys, thereby suppressing the renin-angiotensin mechanism. The combined effect of these actions is a decrease in blood pressure.

The Renin-Angiotensin system, in contrast, is activated in response to low blood pressure to increase it. The two systems work in opposite, or antagonistic, ways to maintain blood pressure homeostasis.

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156. (3)

The liver plays a major excretory role. It breaks down various substances, including bilirubin, a byproduct of hemoglobin breakdown, degraded steroid hormones, and cholesterol. These are then secreted into the bile, which passes into the small intestine and is eventually eliminated in the feces.

The primary role of sweat glands is to help the body cool down through evaporation. While they do excrete small amounts of waste products like urea and salts, their primary role is thermoregulation, and their secondary role is excretion.

Sebaceous glands secrete an oily substance called sebum. Sebum primarily lubricates the skin and hair, but it also contains excretory substances such as sterols, hydrocarbons, and waxes, which are eliminated from the body.

While the kidneys are the main excretory organs, small quantities of nitrogenous wastes like urea and uric acid can indeed be found and eliminated through other body fluids, including saliva, tears, and even milk.

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157. (2)

- While some marine fish are ammonotelic, many (especially cartilaginous fish like sharks) are ureotelic to help with osmoregulation in the hypertonic seawater.
- Terrestrial amphibians (like adult frogs) are primarily ureotelic.
- Most aquatic bony fishes are ammonotelic as they have ample water to excrete toxic ammonia directly.
- Reptiles secrete uric acid as their chief nitrogenous waste. This is an adaptation to terrestrial life where water conservation is important.

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158. (3)

- The peritubular capillaries serve the tubules of all nephrons (cortical and juxtamedullary), whereas the highly specialized *vasa recta* is a key feature only of the juxtamedullary nephrons, crucial for the counter-current exchange mechanism.
- Urine flows from the pyramids into the minor and then major calyces, which all drain into the central renal pelvis before exiting via the ureter.

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159. (2)

- Renal Artery: This vessel brings unfiltered, "dirty" blood to the kidney from the general circulation. It has a high concentration of urea that needs to be removed.
- Renal Vein: This vessel carries filtered, "cleansed" blood away from the kidney after the nephrons have done their job. Since most of the urea has been filtered out and is now part of the urine, the blood in the renal vein has the lowest concentration of urea.
- Abdominal Aorta: This is the major artery that supplies the renal artery. Its urea concentration is high, essentially the same as the renal artery.
- Inferior Vena Cava: The renal vein drains into this larger vein. While the blood here has a low urea concentration, it's a mix of blood from the kidneys and other parts of the lower body, so the blood exclusively in the renal vein is the "cleanest" and has the absolute minimum urea level.

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160. (3)

- Ketonuria, along with glycosuria (glucose in urine), is a classic diagnostic sign of uncontrolled diabetes mellitus.
- The primary role of insulin is to facilitate glucose uptake by cells. Without it, cells are starved for their main fuel source (glucose) despite high levels of it in the blood. The body compensates by breaking down fat reserves (lipolysis) at a high rate. The liver then converts the resulting excess fatty acids into ketone bodies (ketogenesis).

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161. (3)

- The 180 liters of fluid that pass through the glomeruli into the renal tubules is called the glomerular filtrate.
- This filtrate contains water, salts, glucose, urea, and other small molecules. As it flows through the various parts of the renal tubule (PCT, Loop of Henle, DCT, and collecting duct), the body reclaims what it needs.
- The most crucial process is tubular reabsorption, where about 99% of the filtered water, along with essential solutes like glucose, amino acids, and most of the salts, are transported out of the tubule and back into the blood of the peritubular capillaries and *vasa recta*.
- This leaves behind a small volume of fluid containing concentrated metabolic wastes, which becomes the urine that is eventually excreted.

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162. (3)

- Aldosterone, a hormone from the adrenal cortex, acts on the cells of the distal convoluted tubule (DCT) and the collecting duct. Its primary function is to increase the synthesis of pumps and channels that actively reabsorb sodium ions (Na^+) from the filtrate back into the bloodstream. Water then follows the Na^+ by osmosis to maintain the osmotic balance.
- Osmoregulation is indeed a fundamental physiological process. It refers to the homeostatic maintenance of the internal water and solute concentration. Organisms actively regulate the movement of water and ions to ensure that their cells function in a stable environment.

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163. (2)

When a healthy person drinks a large volume of isotonic saline, the body's primary response is to restore normal blood volume and blood pressure. The large volume of fluid intake leads to an increase in blood volume, which is detected by volume receptors and stretch receptors in the heart. This triggers two main responses:

1. Increased Urine Volume: The release of Atrial Natriuretic Factor (ANF) from the heart and the suppression of Antidiuretic Hormone (ADH) from the pituitary gland cause the kidneys to excrete more water, leading to a significant increase in urine volume.
2. Increased Na^+ Concentration: Although the saline is isotonic, the body's overall sodium load has increased. The kidneys, stimulated by the same mechanisms and by the increased filtered load of sodium, will excrete the excess Na^+ to maintain proper electrolyte balance. Since a large volume of water is also being excreted, the total amount of Na^+ in the urine is higher, resulting in a higher concentration compared to the normal physiological state.

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164. (3)

List-I	List-II
Urine	Has a characteristic odour
Brush bordered epithelium	PCT
Glomerulus	A tuft of capillaries
Afferent arteriole	Fine branch of renal artery

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165. (1)

The adrenal glands (A) are located superior to the kidneys (suprarenal means "above the kidney"). The adrenal cortex, an outer layer of the gland, secretes several hormones, including aldosterone, which is crucial for regulating blood pressure and electrolyte balance.

While, B is the renal pelvis, it is a funnel-shaped structure that collects urine from the renal calyces. Ultrafiltration occurs in the glomerulus (part of the nephron), which is located in the renal

C is the medulla, but the Malpighian corpuscles are located in the cortex. The selective reabsorption of glucose and amino acids occurs primarily in the Proximal Convoluted Tubule (PCT), which is also located in the cortex.

D is the cortex, which is the outermost region of the kidney, not the innermost. The medulla is the innermost region and has a striated appearance due to the renal pyramids and their contained collecting ducts.

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166. (1)

The juxtaglomerular apparatus (JGA) is a microscopic structure located at the point where the distal convoluted tubule (DCT) loops back to touch the afferent arteriole of its own glomerulus. It is composed of two main cell types listed here:

- **Juxtaglomerular (JG) cells:** These are modified smooth muscle cells in the wall of the afferent arteriole. They synthesize, store, and secrete the enzyme renin.
- **Macula densa:** These are specialized, densely packed epithelial cells in the wall of the distal convoluted tubule. They act as chemoreceptors that monitor the NaCl concentration of the filtrate.

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167. (1)

Angiotensin II is known to constrict the efferent arteriole (the blood vessel leaving the glomerulus). This increases the pressure in the glomerulus, which helps to maintain the Glomerular Filtration Rate (GFR) even if there is a drop in systemic blood pressure.

By causing blood vessels throughout the body to constrict, it increases peripheral resistance, which leads to a rise in systemic blood pressure.

Angiotensin II acts on the cells of the proximal convoluted tubule (PCT), stimulating the reabsorption of Na⁺ and water. This helps to increase blood volume.

Angiotensin II primary function is to increase blood volume and pressure by vasoconstriction and by stimulating the release of aldosterone, which further enhances Na⁺ reabsorption.

Angiotensin II is not secreted by JG cells. The juxtaglomerular (JG) cells are responsible for secreting renin, which is the enzyme that initiates the conversion of angiotensinogen to angiotensin I, a precursor to Angiotensin II.

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168. (1)

The distinct locations for each step:

- **Ultrafiltration (Glomerular Filtration):** This is a non-selective process that happens only in the renal corpuscle (glomerulus and Bowman's capsule).
- **Tubular Reabsorption:** This is the process of moving useful substances from the filtrate back into the blood. It is a major process that occurs all along the renal tubule (PCT, Loop of Henle, DCT) and the collecting duct.

- **Tubular Secretion:** This is the process of actively moving waste products from the blood into the filtrate. It occurs mainly in the PCT and DCT.

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169. (1)

The kidneys receive a disproportionately large amount of blood for their size because their primary function is to filter the blood. Out of the entire volume of blood pumped by the heart each minute (cardiac output), about one-fifth (20-25%) is diverted to the kidneys for filtration. This is the renal blood flow. It is much larger than the glomerular filtration rate (GFR), which is only the fraction of plasma that gets filtered (~125 ml/min).

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170. (3)

Transitional epithelium is a type of stratified epithelium that can stretch significantly. It is present in parts of the urinary system that need to accommodate changes in volume, such as the ureters, urinary bladder, and a part of the urethra. It is not present in the kidneys or the nephrons, which are lined by different types of epithelia (e.g., simple squamous and simple cuboidal epithelium).

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171. (2)

List-I	List-II
Zones of kidney	Inner medulla and outer cortex
Loop of Henle	Hairpin shaped
<i>Vasa recta</i>	'U' shaped
HCO ₃ ⁻	Reabsorbed in PCT and DCT

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172. (2)

The PCT is the site of bulk reabsorption, where approximately 100% of essential organic nutrients like glucose and amino acids are reabsorbed from the filtrate.

Uric acid has very low solubility in water. This is why it can crystallize and precipitate in joints (causing gout) or in the urinary tract (forming uric acid kidney stones).

The renal pyramids make up the renal medulla and are primarily composed of the parallel arrangement of the loops of Henle and the collecting ducts.

The convoluted tubules (both PCT and DCT) are located in the outer region of the kidney, the renal cortex. The renal pyramids are in the medulla.

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173. (1)

A nephron consists of two main parts: a renal corpuscle and a renal tubule. Since every single nephron has exactly one renal corpuscle, the total count of nephrons must be equal to the total count of renal corpuscles.

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174. (1)

The filtration of blood in the glomerulus is a precise process that occurs across a three-layer barrier. The correct sequence is from the inside of the blood vessel outwards to the space of the Bowman's capsule:

1. **Fenestrated Endothelium:** This is the wall of the glomerular capillary. It has large pores (fenestrations) that allow water, solutes, and plasma proteins to pass through but block blood cells.
2. **Basement Membrane:** This is a non-cellular layer of extracellular matrix that lies between the endothelium and the podocytes. It acts as a size-selective and charge-selective barrier, preventing large plasma proteins (like albumin) from passing through.
3. **Filtration Slits of the Podocytes:** Podocytes are specialized cells that form the visceral (inner) layer of the Bowman's capsule. They have foot-like extensions called pedicels that wrap around the capillaries. The narrow gaps between these pedicels are called filtration slits, and they provide the final layer of filtration, preventing any medium-sized proteins from passing into the filtrate.

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175. (1)

List-I	List-II
Platyhelminthes	Flame cells
Vertebrates	Complex tubular forms of excretory structures
Earthworm	Nephridia
Renal capsule	Tough protective covering around kidney

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176. (2)

Diabetes Insipidus is a disorder characterized by the inability of the kidneys to conserve water.

- The Cause: It is caused by an insufficient production or release of Antidiuretic Hormone (ADH) from the posterior pituitary (known as Central DI) or the failure of the kidney's collecting ducts to respond to ADH (known as Nephrogenic DI).

- The Effect: ADH's normal function is to make the collecting ducts permeable to water, allowing water to be reabsorbed back into the body. Without sufficient ADH action, this reabsorption does not happen, and a massive amount of water is lost. This results in the excretion of a very large volume of dilute, tasteless ("insipid") urine, leading to dehydration and intense thirst (polydipsia).

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177. (3)

- Renin is an enzyme, not a hormone. It is secreted by the juxtaglomerular cells of the kidney and acts on angiotensinogen to initiate the renin-angiotensin-aldosterone system.
- Urochrome, also spelled urobilin, is a yellowish pigment formed from the breakdown of hemoglobin, which gives urine its characteristic yellow color.
- Diuresis refers to the increased or excessive production of urine, which may or may not be accompanied by frequent urination. The medical term for frequent urination is polyuria.
- Urea is the primary nitrogenous waste product in humans, formed from the breakdown of amino acids from proteins. A lower protein intake means fewer amino acids are available for metabolism, thus less urea is produced and excreted.
- Renal calculi, or kidney stones, are hard, crystalline mineral deposits that form in the kidney. They are primarily composed of calcium oxalate, uric acid, or other salts.

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178. (1)

The Malpighian corpuscle, also known as the renal corpuscle, is the filtering unit of the nephron. It is indeed composed of two main parts: the glomerulus, a tuft of capillaries, and the Bowman's capsule, a double-walled, cup-like structure that surrounds the glomerulus.

The surgical removal of one kidney, or a unilateral nephrectomy, is not fatal in a healthy individual. The remaining kidney is fully capable of taking over the functions of both kidneys and can undergo hypertrophy (an increase in size) to compensate. While there might be some short-term adjustments, a person can live a normal, healthy life with just one kidney. Fatal renal failure occurs when both kidneys fail, a condition known as bilateral renal failure.

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179. (1)

Ultrafiltration is a passive process that relies on a pressure gradient to force water and small solutes out of the blood and into the Bowman's capsule. The main force pushing the fluid out is the high blood pressure within the glomerular capillaries, known as the Glomerular Blood Hydrostatic Pressure (GBHP).

The afferent arteriole is wider, allowing a large volume of blood to enter the glomerulus easily. The efferent arteriole is narrower, creating a bottleneck that resists the outflow of blood.

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180. (3)

The body's metabolic processes (especially protein metabolism) produce a net excess of acid. To maintain the blood's tightly regulated pH (around 7.4), the kidneys must excrete this excess acid. They do this primarily through:



- **Tubular Secretion of H^+ :** The cells of the renal tubules (mainly the PCT, DCT, and collecting ducts) actively pump hydrogen ions from the blood into the filtrate.
- **Reabsorption of HCO_3^- :** The same tubular cells reabsorb bicarbonate ions (a base) from the filtrate and return them to the blood. This acts to conserve the body's primary buffer.

The combination of adding an acid (H^+) to the filtrate and removing a base (HCO_3^-) from it is what makes the resulting urine acidic.

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